Graphura Social Media Content Intelligence

Machine Learning Project Report

1. Executive Summary

This project builds a **Machine Learning–driven Content Intelligence System** for Graphura to **predict social media post performance** and support **data-driven content planning**.

The system analyzes historical Instagram and LinkedIn post data, classifies content into **High, Medium, or Low performance**, and connects model predictions with **visual dashboards** for actionable insights.

Key Outcome:

The best-performing ML models achieved **~94% accuracy** in predicting post performance, enabling Graphura to optimize content type, format, and engagement strategy before posting.

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2. Business Problem Statement

Graphura needed answers to:

- Which **post types** perform best on Instagram and LinkedIn?
- What **engagement signals** drive high performance?
- Can we **predict post success in advance** instead of relying on intuition?

This project solves these problems by combining:

- **Exploratory dashboards** (what happened)
- **Machine Learning models** (what will happen)

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3. Data Overview

- **Total Records:** 175 social media posts
- **Platforms Covered:** Instagram, LinkedIn
- **Key Features Used:**
    - Views, Reach, Likes
    - Shares, Comments, Saves, Follows

- Post Type (Image, Carousel, Reel, Video, Document)
- Hashtag Count

Missing values were handled using **mean/mode imputation**, and irrelevant columns were removed during preprocessing .

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4. Feature Engineering & Target Variable

Engagement Metrics Created

- **Engagement Rate by Reach (%)**
- **Engagement Rate by Views (%)**
- **Average Engagement Rate (%)**

Performance Classification Logic

Posts were labeled as:

- **Low Performance**
- **Medium Performance**
- **High Performance**

Based on engagement rate thresholds derived from data distribution.

This converted a raw analytics problem into a **multi-class classification ML problem**.

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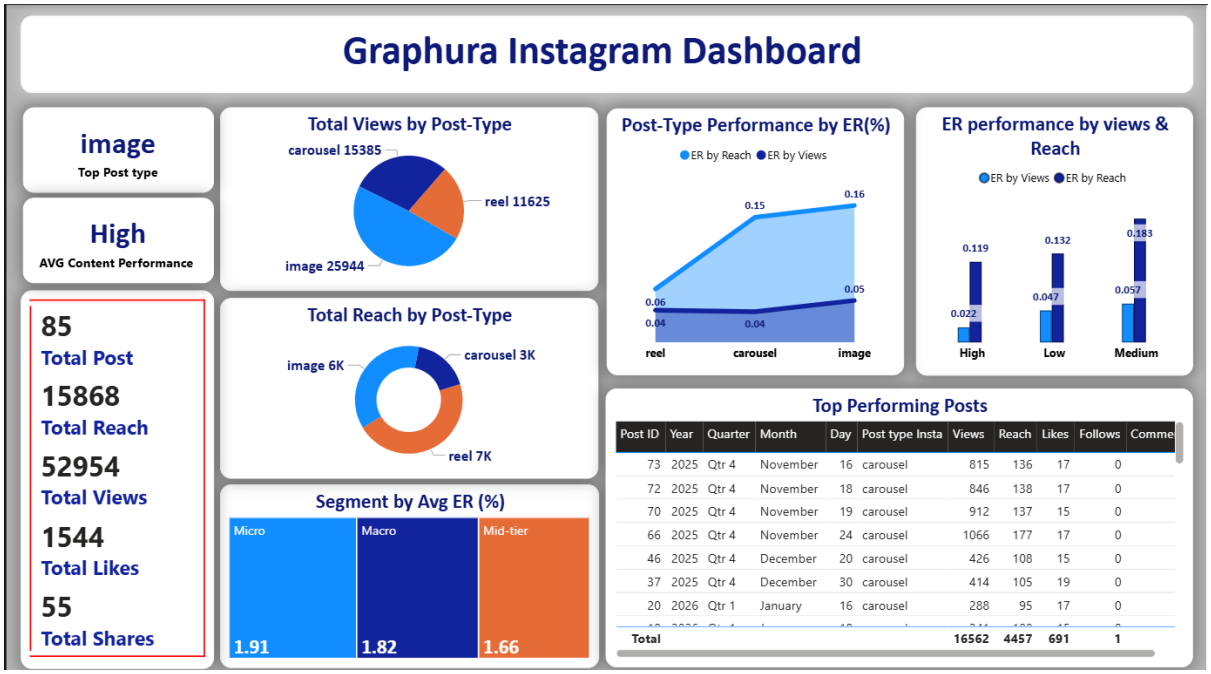
5. Dashboard Insights (What the Data Shows)

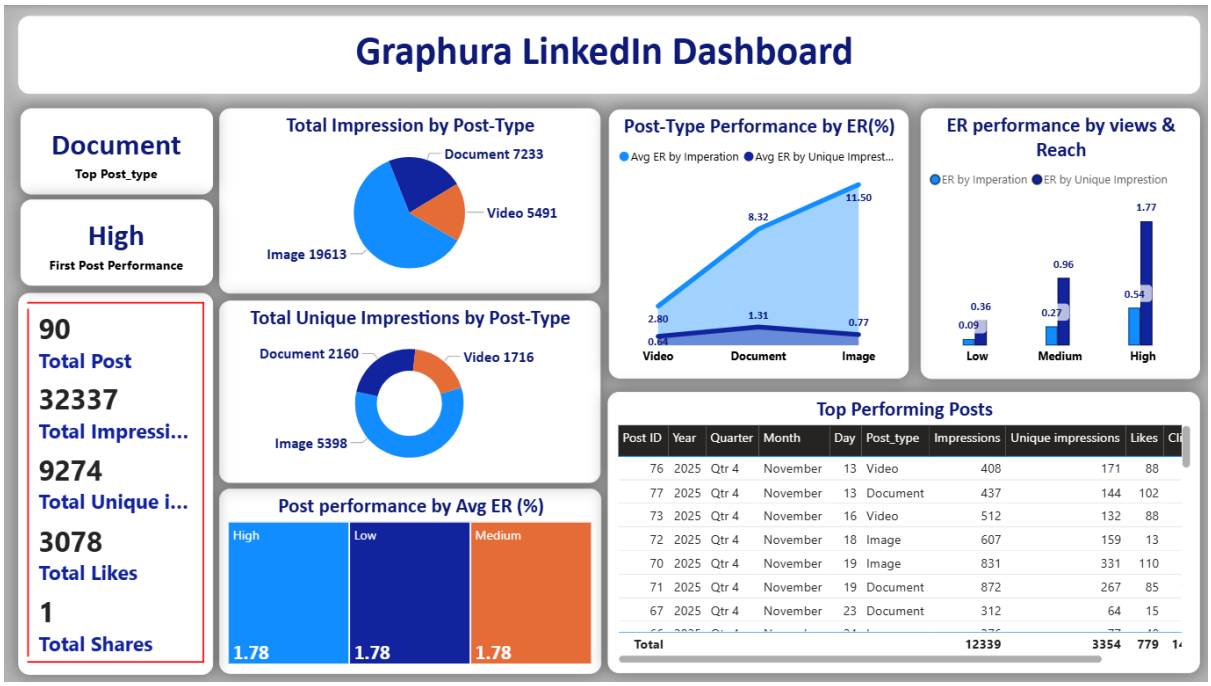


📸 Instagram Dashboard Insights

- **Top Performing Post Type:** Image
- **Highest Views & Reach:** Image & Carousel
- **Engagement Rate:**
  - Images perform best in ER by reach
  - Medium-performing content shows higher ER consistency
- **Total Stats:**
  - 85 posts
  - 52,954 total views
  - 15,868 total reach

Insight: *Static images and carousels consistently outperform reels for Graphura's audience.*

LinkedIn Dashboard Insights



- **Top Post Type:** Document
- **Highest Impressions:** Image posts
- **Best ER:** High-performance content across all formats
- **Total Stats:**
  - 90 posts
  - 32,337 impressions
  - 9,274 unique impressions

Insight: *Educational/document-style content works best on LinkedIn.*

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6. Machine Learning Pipeline

Preprocessing

- Numerical features → **RobustScaler**
- Categorical features → **OneHotEncoding**
- Unified using **ColumnTransformer + Pipeline**

This ensured clean, leakage-free training.

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7. Models Trained & Evaluated

Model	Accuracy
Logistic Regression	**94.28%**
KNN	**94.28%**
Random Forest	**94.28%**
Gradient Boosting	**94.28%**
Decision Tree	91.42%
SVM	88.57%
Naive Bayes	77.14%

Final Model Selected: Random Forest
(Balanced performance + feature interpretability)

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8. Feature Importance (Key Drivers)

From the Random Forest model:

Top influencing features:

1. Likes
2. Comments
3. Reach
4. Views
5. Post Type (Video/Image)
6. Saves
7. Hashtag Count

Business takeaway:

Engagement depth (likes + comments) matters more than just reach.

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9. Model Validation (Manual Testing)

Three real-world test cases were validated:

- Low engagement → **Predicted Low**
- Average engagement → **Predicted Medium**
- Viral content → **Predicted High**

Predictions matched expected outcomes, confirming model reliability .

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10. Deployment Readiness

- Model saved as: best_model_pipeline.joblib
- Handles **missing inputs safely**
- Ready for:
    - API integration
    - Dashboard embedding
    - Content recommendation systems

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11. Business Impact

Predict post performance **before publishing**
Optimize content format per platform
Reduce trial-and-error posting
Support marketing decisions with data

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12. Future Enhancements

- Add **posting time & weekday**
- Include **caption sentiment analysis**
- Platform-specific models
- Real-time prediction dashboard